Practical 4: Jetty Server for WAR Application

Step 1: Check Java and Maven

java -version

mvn -version

Step 2: Create Maven Web Project

mvn archetype:generate -DgroupId=com.example -DartifactId=my-webapp -DarchetypeArtifactId=maven-archetype-webapp -DarchetypeVersion=1.4 -DinteractiveMode=false

Step 3: Move into Project Folder

cd my-webapp

Step 4: Edit index.jsp

Path: src/main/webapp/index.jsp

```
<html>

<body>

<h1>Jetty Server is running this Maven Web App!</h1>

</body>

</html>
```

Step 5: Edit pom.xml (Jetty Plugin configuration)

Step 6: Build Project

mvn clean package

Step 7: Run Jetty Server

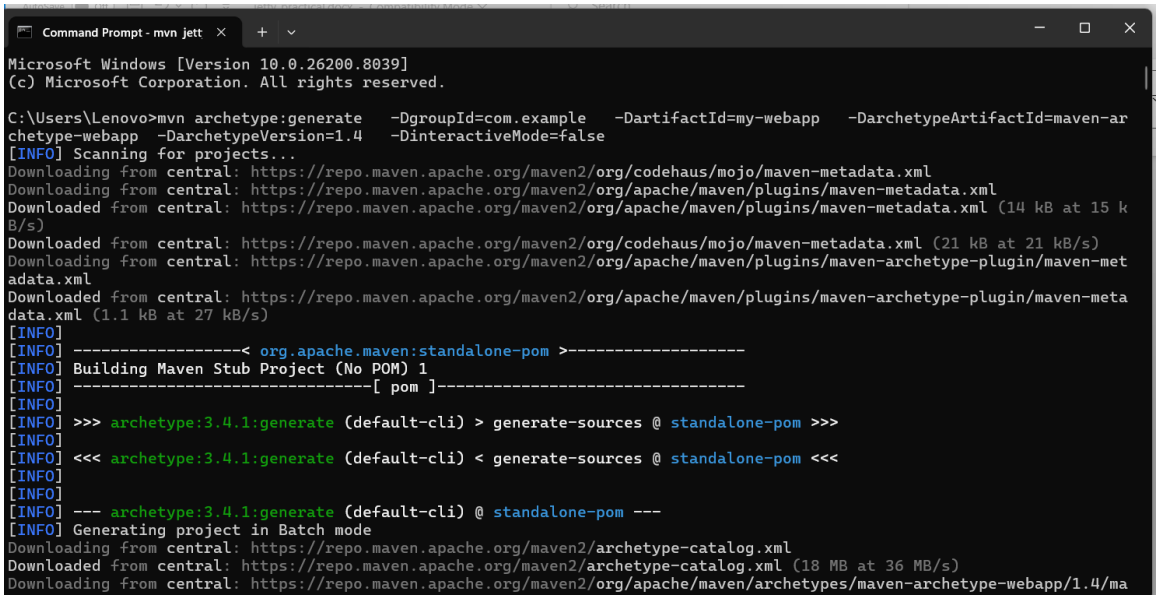
mvn jetty:run-war

Step 8: Open Browser

http://localhost:8081/my-webapp

Output:

Jetty Server is running this Maven Web App!



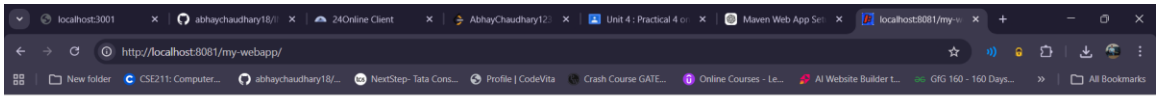
```
Command Prompt - mvn jett
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Lenovo>mvn archetype:generate   -DgroupId=com.example   -DartifactId=my-webapp   -DarchetypeArtifactId=maven-ar
chetype-webapp  -DarchetypeVersion=1.4   -DinteractiveMode=false
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 15 k
B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 21 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-met
adata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-meta
data.xml (1.1 kB at 27 kB/s)
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 36 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.4/ma
```

```
Command Prompt - mvn jett
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 36 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.4/ma
ven-archetype-webapp-1.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.4/mav
en-archetype-webapp-1.4.jar (6.8 kB at 131 kB/s)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Archetype: maven-archetype-webapp:1.4
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: my-webapp
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: packageInPathFormat, Value: com/example
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: my-webapp
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Project created from Archetype in dir: C:\Users\Lenovo\my-webapp
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.308 s
[INFO] Finished at: 2026-04-01T01:48:09+05:30
[INFO] ------------------------------------------------------------------------
```

```
Command Prompt - mvn jett

C:\Users\Lenovo>cd my-webapp

C:\Users\Lenovo\my-webapp>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:my-webapp >------------------------
[INFO] Building my-webapp Maven Webapp 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ war ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.2.2/maven-war
-plugin-3.2.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.2.2/maven-war-
plugin-3.2.2.pom (9.7 kB at 11 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.2.2/maven-war
-plugin-3.2.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-war-plugin/3.2.2/maven-war-
plugin-3.2.2.jar (91 kB at 773 kB/s)
[INFO]
[INFO] --- clean:3.1.0:clean (default-clean) @ my-webapp ---
[INFO]
[INFO] --- resources:3.0.2:resources (default-resources) @ my-webapp ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory C:\Users\Lenovo\my-webapp\src\main\resources
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ my-webapp ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.0.2:testResources (default-testResources) @ my-webapp ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
```

```
Command Prompt - mvn jett
s-1.16.1.jar (560 kB at 1.8 MB/s)
[INFO] Packaging webapp
[INFO] Assembling webapp [my-webapp] in [C:\Users\Lenovo\my-webapp\target\my-webapp]
[INFO] Processing war project
[INFO] Copying webapp resources [C:\Users\Lenovo\my-webapp\src\main\webapp]
[INFO] Webapp assembled in [157 msecs]
[INFO] Building war: C:\Users\Lenovo\my-webapp\target\my-webapp.war
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  9.581 s
[INFO] Finished at: 2026-04-01T01:49:19+05:30
[INFO] ------------------------------------------------------------------------

C:\Users\Lenovo\my-webapp>notepad pom.xml

C:\Users\Lenovo\my-webapp>mvn jetty:run-war
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-maven-plugin/9.4.54.v20240208/jet
ty-maven-plugin-9.4.54.v20240208.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-maven-plugin/9.4.54.v20240208/jett
y-maven-plugin-9.4.54.v20240208.pom (12 kB at 16 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-project/9.4.54.v20240208/jetty-pr
oject-9.4.54.v20240208.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-project/9.4.54.v20240208/jetty-pro
ject-9.4.54.v20240208.pom (73 kB at 864 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/testcontainers/testcontainers-bom/1.16.1/testcontaine
rs-bom-1.16.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/testcontainers/testcontainers-bom/1.16.1/testcontainer
s-bom-1.16.1.pom (7.2 kB at 154 kB/s)
```

```
Command Prompt - mvn jett
Downloading from central: https://repo.maven.apache.org/maven2/javax/transaction/javax.transaction-api/1.3/javax.transac
tion-api-1.3.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/taglibs/taglibs-standard-spec/1.2.5/taglibs-sta
ndard-spec-1.2.5.jar (40 kB at 20 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/mortbay/jasper/apache-jsp/8.5.70/apache-jsp-8.5.70.jar
 (1.1 MB at 510 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/javax/transaction/javax.transaction-api/1.3/javax.transact
ion-api-1.3.jar (28 kB at 14 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/taglibs/taglibs-standard-impl/1.2.5/taglibs-sta
ndard-impl-1.2.5.jar (206 kB at 99 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jdt/ecj/3.19.0/ecj-3.19.0.jar (2.9 MB at 1.3 M
B/s)
[INFO] Logging initialized @24858ms to org.eclipse.jetty.util.log.Slf4jLog
[INFO] Configuring Jetty for project: my-webapp
[INFO] Context path = /my-webapp
[INFO] Tmp directory = C:\Users\Lenovo\my-webapp\target\tmp
[INFO] Web defaults = org/eclipse/jetty/webapp/webdefault.xml
[INFO] Web overrides =  none
[INFO] jetty-9.4.54.v20240208; built: 2024-02-08T19:42:39.027Z; git: cef3fbd6d736a21e7d541a5db490381d95a2047d; jvm 17.0.
12+8-LTS-286
[INFO] Scanning elapsed time=114ms
[INFO] DefaultSessionIdManager workerName=node0
[INFO] No SessionScavenger set, using defaults
[INFO] node0 Scavenging every 660000ms
[INFO] Started o.e.j.m.p.JettyWebAppContext@4b09d1c3{Archetype Created Web Application,/my-webapp,file:///C:/Users/Lenov
o/my-webapp/target/my-webapp-1.0-SNAPSHOT/,AVAILABLE}{C:\Users\Lenovo\my-webapp\target\my-webapp-1.0-SNAPSHOT.war}
[INFO] Started ServerConnector@58a8ea6f{HTTP/1.1, (http/1.1)}{0.0.0.0:8081}
[INFO] Started @26688ms
[INFO] Started Jetty Server
```


localhost:3001
24Online Client
Unit 4 : Practical 4
Maven Web App Ser
localhost:8081/my-w
http://localhost:8081/my-webapp/
New folder
CSE211: Computer...
NextStep- Tata Cons...
Profile | CodeVita
Crash Course GATE...
Online Courses - Le...
AI Website Builder t...
GfG 160 - 160 Days...
All Bookmarks

Hello World!